DAY 1 · TODAY

Neural network foundations

- Tensors and shapes
- Neuron, perceptron
- MLPs, activations
- Losses, gradients
- Backpropagation

Check a forward and backward pass

DAY 2

Training neural networks reliably

- Learning curves
- Initialization
- Normalization
- Optimizers
- Regularization

Reproduce a training experiment

DAY 3

Energy modelling capstone

- Data audits
- ESP gas lock
- PE log curve
- Sensor signals
- Error analysis

Defend a model comparison

DAY 4

Convolutional neural networks

- Convolution, pooling
- Architectures
- Transfer learning
- Segmentation
- Seismic, borehole

Check an image or signal model

DAY 5

Sequence models and attention

- Windows, baselines
- RNN, LSTM, TCN
- Attention
- Transformers
- Forecast intervals

Evaluate a future forecast